

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **6562**

L 61_479_986_L 1505_48302

AY: 2025-26	Session: Odd MSE October 2025	Date: 08-10-2025
Sem: 1	Slot: 10:00am - 11:00am	C. Code: CS413
Exam Type: Written	Block: 202	Seat No: 2022600022
Course Section: 1		



Name of the candidate. Aman Kangila

Candidate's UID number:

2	0	2	2	6	0	0	0	2	2
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Examination class room number:

2	0	2
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Invigilator's signature with date:

	08/10/2025
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(To be filled in by the candidate)

EXAMINATION: BE Btech MSE
(For example FY MCA/FY BTECH)

SEMESTER: VII

PROGRAM: CSE(AI-ML)

I / II / III / IV / V / VI / VII / VIII ✓

DAY AND DATE: 8th Oct 2025 - Wednesday

COURSE NAME: DL

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Q1) a) $f(w) = \frac{1}{2} \|xw - y\|^2$

$$\frac{\partial f(w)}{\partial w} = \frac{\partial}{\partial w} \left(\frac{1}{2} \|xw - y\|^2 \right)$$

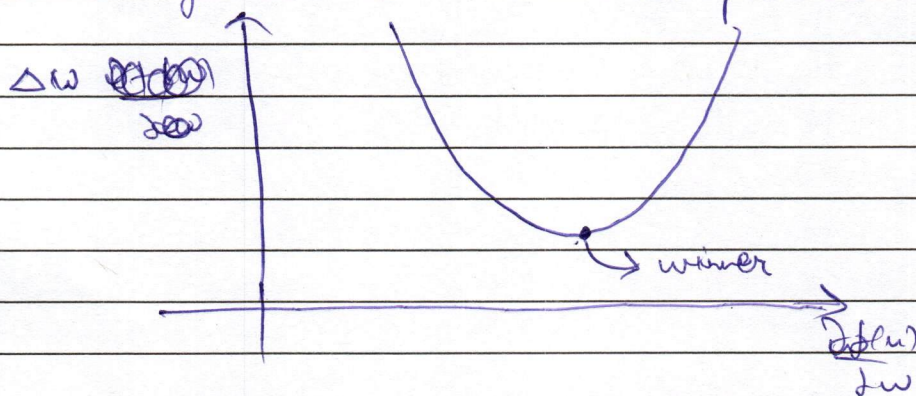
$$\frac{\partial f(w)}{\partial w} = \frac{1}{2} \times \frac{\partial (\frac{1}{2} \|xw - y\|^2)}{\partial \|xw - y\|} \times \frac{\partial \|xw - y\|}{\partial w}$$

$$\frac{\partial f(w)}{\partial w} = \frac{1}{2} \times 2 (\|xw - y\|) \times \frac{\partial \|xw - y\|}{\partial w}$$

$$\frac{\partial f(w)}{\partial w} = \|xw - y\| \times (x)$$

$$\boxed{\frac{\partial f(w)}{\partial w} = (x) \|xw - y\|}$$

Gradient descent can be observed from the following graph to produce a convergence at a certain point:



$$\boxed{\text{We know } \Delta w = -\eta \nabla E(w)}$$

Sr. No. 6562



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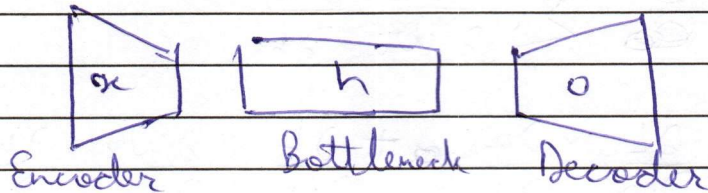
Q1) Overfitting occurs when model capacity is too high because the model tends to learn the inputs owing to its capability to do so, thus overfitting.

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	Q2)a)	In the case of a standard 2D convolution, the number of a multiplications, or the computational complexity will be: $(H-F+1) \times (H-F+1) \times C \times K$ Operations due to the convolution operation by the filter Since its 2D convolution, the third dimension will be treated as number of layers As K filters are present, the convolution will occur K times further. In the case of a depthwise separable convolution, the computational complexity will decrease, as the filter being used will be made of size $F \times F \times C$ $F \times F \times (H/W/C)$ with K filters. Any one of the input dimensions can be taken as the third dimension of the filter, giving us the complexity as: $(H-F+1) \times (H-F+1) \times K$ This Therefore Depthwise separable convolutions are critical in modern architecture owing to the increase in input dimensionalities. For example, in the case of a 4K RGB image as an input, the pixel input would be $2048 \times 2048 \times 3$, which when passed through the feed forward network, would require $3 \times 2048 \times 2048 \times 2048$ computation due to the hidden layers. Thus depthwise separable convolutions drastically reduce parameters by increasing dimensions of filters.

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Q2) f) An ~~an~~ under complete autoencoder has the bottleneck or latent space dimension lesser than the input dimensions.



and for Under complete autoencoder,
 $\dim(h) < \dim(x)$

Therefore, we observe ~~an~~ dimensionality reduction for the latent space. This ~~to~~ reduces computation for the model.

On the otherhand, an over complete autoencoder has the bottleneck dimensions greater than or equal to the input dimensions,

$$\dim(h) \geq \dim(x)$$

Regularization is needed in ~~A~~ Over Complete Autoencoders, otherwise the ~~latent~~ ~~latent~~ model might simply just copy the input and pad it over to match the dimensions, thus learning the input to overfitting. Thus it learns the identity of the input, and simply ~~converts~~ ~~converts~~ reconverts it back to the input state at the decoder.

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Q3a) The recursive gradient function for an RNN is

$$\frac{\partial L}{\partial h_{t+k}} = \frac{\partial L}{\partial h_t} \sum_{i=1}^k \frac{\partial L}{\partial w_i} \frac{\partial w_i}{\partial h_t}$$

$$= \frac{\partial L}{\partial w_k} \frac{\partial w_k}{\partial h_t} + \frac{\partial L}{\partial w_k} \frac{\partial w_k}{\partial w_{k-1}} \frac{\partial w_{k-1}}{\partial h_{t-1}} +$$

$$\frac{\partial L}{\partial w_k} \frac{\partial w_k}{\partial w_{k-1}} \frac{\partial w_{k-1}}{\partial w_{k-2}} \frac{\partial w_{k-2}}{\partial h_{t-2}} + \dots +$$

$$\frac{\partial L}{\partial w_k} \frac{\partial w_k}{\partial w_{k-1}} \dots \frac{\partial w_1}{\partial h_t}$$

~~Here we~~

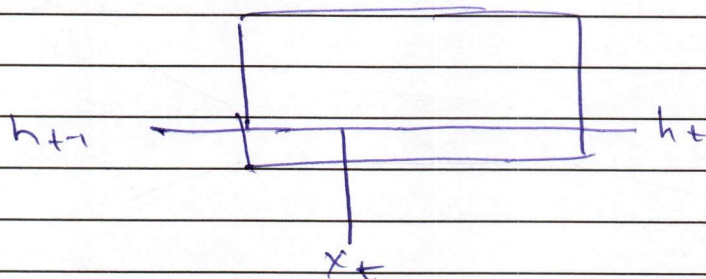
Here we can see that the initial inputs do not impact the learnings over time, as their gradient becomes lesser and lesser, giving rise to the Vanishing Gradient Problem.

This can be fixed using an LSTM, which consists of a Input gate, Forget gate and an Output gate.

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Q3) f) GRU consists of ~~the~~ just the Update gate and Reset gate.



~~$\tilde{C}_t = \tanh(W_c h_{t+1} + W_x X_t + b_c)$~~

~~$C_t = U_t \tilde{C}_t + (1 - U_t) X_t$~~

~~$\tilde{C}_t = \tanh(\dots)$~~

$$C_t = U_t \tilde{C}_t + (1 - U_t) X_t$$

$$\tilde{C}_t = \tanh(W_c h_{t+1} + W_x X_t + b_c)$$

$$U_t = \sigma(W_u h_{t+1} + W_n X_t + b_u)$$

~~$h_t = \text{softmax}(C_t X_t)$~~

$$h_t = \text{softmax}(C_t X_t)$$

GRUs ~~are~~ have fewer parameters than LSTM, as GRU only consists of the Update gate and the Reset gate, whereas LSTM consists of the Input gate, Forget gate and the Output gate.

Therefore in case where the data is large, or more complex, GRUs tend to be better than LSTMs.

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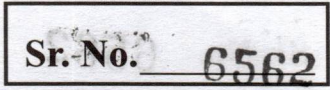


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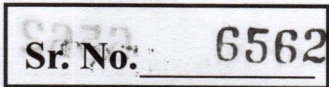
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Rough work

$$\Delta w = -\eta \nabla E(w)$$

$$g = -\eta x (xw - y)$$

$$xw = y$$

$$xx^T = I$$

$$1 - \eta (xw - y)$$

$$xx^T - \lambda I = 0$$

$$I - \lambda I = 0$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \approx 0$$

$$\frac{1}{xw - y} = \lambda$$

$$1 - \lambda = 0$$

$$\lambda = 1$$

$$1 - \lambda$$